

TYRONE XU

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EDUCATION

B.S. Mechanical Engineering — Automotive Focus

Expected Dec 2027

Georgia Institute of Technology, Atlanta, GA | Zell Miller Scholarship

Relevant Coursework: Dynamics of Rigid Bodies, Materials Engineering, Design & Fabrication, Deformable Bodies, Fluid Mechanics, Thermodynamics, Circuits and Electronics

TECHNICAL SKILLS

CAD / Simulation: Fusion 360, SolidWorks, SolidWorks Simulation/Flow Simulation, MATLAB

Programming: Python, C/C++, JavaScript, HTML

Manufacturing & Fabrication: CNC Machining, Mill, Lathe, Waterjet, Laser Cutting, 3D Printing, Welding

Diagnostics & Testing: OBD Scanners, Data Logging Software, Dyno Tuning, Engine Performance Analysis

EXPERIENCE

Powertrain & Aerodynamics Engineer | Wreck Racing — Georgia Tech Motorsports Club

Aug 2023 – Present

Designed, fabricated, and validated a fully custom race vehicle from the ground up under a \$2,000 budget constraint.

- Designed and validated suspension A-arms in Fusion 360 to improve front/rear rigidity and steering response on a custom tube chassis.
- Designed and fabricated custom engine mounts for a 4.2L Audi V8 swap into a Dodge Conquest, resolving drivetrain alignment, clearance, and chassis mounting constraints.
- Designed and built a custom engine test stand to tune and characterize the V8 before installation, catching calibration issues off vehicle before final assembly.
- Used OBD-II scanners and data logging software to monitor ECU parameters, resolve fault codes, and refine engine calibration during test runs.
- Designed and 3D printed a rear spoiler in Fusion 360; iterated CFD geometry to improve rear downforce and used SolidWorks FEA to verify the mount design under expected loading.

Sales Intern | Vector Marketing

May 2024 – Aug 2024

- Generated \$10,000+ in sales and ranked among the top new representatives in the region.
- Delivered 10+ in-home and online product demos per week while maintaining an 86% close rate.

PROJECTS

Mechanical Design Engineer | ME 2110 Design Competition

Spring 2026

- Competed against 80 teams in a 40-second robotics challenge, placing 2nd overall on a 4-person team.
- Designed and fabricated an automated ball-bearing slider ramp sorting mechanism capable of classifying and routing objects by their geometry.
- Modeled all components in CAD and fabricated the mechanism with 3D-printed parts and laser-cut MDF, keeping the assembly lightweight, low-cost, and deployed consistently.
- Led design reviews and subsystem check-ins to resolve interface conflicts before final assembly.

Mechanical Systems & Powertrain Engineer | 2014 Subaru BRZ Performance & Maintenance Project

2022 – Present

- Diagnosed engine fluid ingress and completed timing cover and oil pan reseals using OEM service procedures.
- Designed a custom VVT solenoid retention bracket after a stripped mounting thread; prototyped in SolidWorks and fabricated the final part with waterjet-cut metal and welding.
- Installed oil pressure and temperature gauges to monitor powertrain health and track baseline data over time.
- Serviced transmission oil, rear differential fluid, coolant, and brake fluid using OEM specifications.
- Upgraded brake rotors, pads, and exhaust system to improve braking response and exhaust flow.

LEADERSHIP & COMMUNITY INVOLVEMENT

Skills Coordinator | Wreck Racing — Georgia Tech Motorsports Club

Spring 2025 – Present

- Manage team access to the Student Competition Center Common Machining Area (CMA), overseeing SUMS onboarding and current CMA training requirements for new members.
- Serve on the Executive Board, coordinating member certifications and shop training standards across CNC, Lathe, Mill, Welding, and fabrication equipment.

Additional Information

Available for internship Fall 2026 and beyond.